

The Improved MAID Model

Multi-Agent Impersonation, Epistemic Uncertainty, and Strategic Market Distortion

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Abstract

We develop an improved version of the Multi-Agent Impersonation with Fuzzy Knowledge (MAID) model for heterogeneous financial and information economies. The original framework considers domestic, international, and adversarial agents that interact through a common market and choose impersonation intensities under incomplete knowledge. The revised model introduces five extensions.

First, the market signal enters the payoff through the agent's impersonation action, making the economic interpretation of the first-order conditions explicit. Second, fuzzy knowledge is treated as a time-varying epistemic state rather than as a fixed scalar. Third, strategic interference is represented by a state-dependent ratio that can evolve with market conditions. Fourth, the constrained equilibrium is formulated as a Karush–Kuhn–Tucker problem rather than by mechanically truncating an unconstrained solution. Fifth, the model is embedded in a state-space observation system in which strategic impersonation can distort observed financial variables.

The resulting framework distinguishes fundamental economic uncertainty, measurement noise, epistemic uncertainty, and strategic observation distortion. It produces a tractable static equilibrium, a dynamic multi-agent extension, an endogenous uncertainty channel, and an empirical route based on generalized method of moments, state-space estimation, and multi-agent reinforcement learning.

The paper ends with “The End”

Contents

1	Introduction	2
2	The Economic Environment	3
2.1	Agents and action spaces	3
2.2	Fundamental market state	3
3	Epistemic Uncertainty	4
3.1	Fuzzy knowledge	4
3.2	Noisy market signals	4
4	Payoff Structure	4
5	Best Responses	5
6	Matrix Representation	5
7	Closed-Form Interior Equilibrium	6
8	Constrained Nash Equilibrium	6
9	Potential Game Representation	7
10	Symmetric Equilibrium	7
11	Strategic Substitutability	8

12 The Knowledge Wedge	8
13 Dynamic MAID Model	8
14 Strategic Observation Distortion	9
15 State-Space Form	9
16 Time-Varying Measurement Uncertainty	10
17 Bayesian Updating of Knowledge	10
18 Variance Propagation	11
19 Empirical Identification	11
20 Financial-Market Application	11
21 Cross-Institutional Measurement	12
22 Strategic Distortion Index	12
23 Regime Classification	13
24 Multi-Agent Reinforcement Learning Extension	13
25 Robustness and Adversarial Learning	14
26 Welfare and Mechanism Design	14
27 Identification Strategy	14
27.1 Stage 1: Fundamental state estimation	14
27.2 Stage 2: Knowledge estimation	15
27.3 Stage 3: Strategic estimation	15
27.4 Stage 4: Observation-distortion estimation	15
27.5 Stage 5: Joint estimation	15
28 Testable Hypotheses	15
28.1 Hypothesis 1: Strategic substitutability	15
28.2 Hypothesis 2: Knowledge advantage	15
28.3 Hypothesis 3: Adversarial knowledge wedge	15
28.4 Hypothesis 4: Epistemic amplification	16
28.5 Hypothesis 5: Strategic measurement distortion	16
29 Limitations	16
30 Conclusion	16

1 Introduction

Financial markets are populated by agents that possess heterogeneous information, interact strategically, and may have incentives to misrepresent their identities, intentions, beliefs, or market positions. In such an environment, an observed market signal need not be a direct representation of the underlying economic state.

The original MAID model formalises this environment using three representative agents:

$$J = \{D, I, A\}, \tag{1}$$

where D denotes a domestic agent, I an international agent, and A an adversarial agent. Each agent possesses imperfect knowledge of a common market state and chooses an impersonation intensity.

The improved model retains this structure but distinguishes four objects:

$$\text{fundamental state,} \quad \text{market signal,} \quad \text{strategic action,} \quad \text{observed signal.} \quad (2)$$

This distinction is important because uncertainty about the fundamental state and distortion of its observation are conceptually different.

The central architecture is therefore

$$X_t \longrightarrow M_t \longrightarrow \widetilde{M}_{j,t} \longrightarrow x_{j,t}^* \longrightarrow Y_t^{obs}. \quad (3)$$

Here X_t is the underlying economic state, M_t is the fundamental market quantity, $\widetilde{M}_{j,t}$ is the noisy signal observed by agent j , $x_{j,t}^*$ is equilibrium impersonation, and Y_t^{obs} is an observable financial variable.

The purpose of the revised model is therefore not merely to solve an impersonation game. It is to provide a mechanism through which strategic interaction and epistemic uncertainty can enter an empirical financial measurement system.

2 The Economic Environment

2.1 Agents and action spaces

Let

$$J = \{D, I, A\}. \quad (4)$$

Each agent $j \in J$ chooses an impersonation level

$$x_{j,t} \in [0, 1]. \quad (5)$$

The interpretation is

$$x_{j,t} = 0 \quad \Longleftrightarrow \quad \text{full authenticity,} \quad (6)$$

and

$$x_{j,t} = 1 \quad \Longleftrightarrow \quad \text{maximum impersonation.} \quad (7)$$

The action vector is

$$\mathbf{x}_t = \begin{pmatrix} x_{D,t} \\ x_{I,t} \\ x_{A,t} \end{pmatrix} \in [0, 1]^3. \quad (8)$$

2.2 Fundamental market state

Let

$$X_t \in \mathbb{R}^n \quad (9)$$

denote the latent economic state.

The fundamental market value is

$$M_t = m(X_t), \quad M_t > 0. \quad (10)$$

For a linear specification,

$$M_t = \mathbf{m}^\top X_t. \quad (11)$$

The important distinction is that M_t is a fundamental quantity whereas the agent does not observe M_t perfectly.

3 Epistemic Uncertainty

3.1 Fuzzy knowledge

Each agent possesses a time-varying knowledge state

$$K_{j,t} \in [0, 1]. \quad (12)$$

The interpretation is

$$K_{j,t} = 1 \iff \text{perfect observation of the relevant market state,} \quad (13)$$

while

$$K_{j,t} = 0 \iff \text{complete epistemic uncertainty.} \quad (14)$$

Define the knowledge vector

$$\mathbf{K}_t = \begin{pmatrix} K_{D,t} \\ K_{I,t} \\ K_{A,t} \end{pmatrix}. \quad (15)$$

3.2 Noisy market signals

Agent j observes

$$\widetilde{M}_{j,t} = K_{j,t}M_t + (1 - K_{j,t})\varepsilon_{j,t}, \quad (16)$$

where

$$\varepsilon_{j,t} \sim U[0, 1]. \quad (17)$$

More generally, the noise may be written as

$$\varepsilon_{j,t} \sim F_{\varepsilon,j,t}, \quad (18)$$

with

$$\text{Var}(\varepsilon_{j,t}) = \sigma_{\varepsilon,j,t}^2. \quad (19)$$

The conditional variance of the signal is therefore

$$\text{Var}(\widetilde{M}_{j,t} \mid M_t, K_{j,t}) = (1 - K_{j,t})^2 \sigma_{\varepsilon,j,t}^2. \quad (20)$$

Thus epistemic uncertainty enters the model directly through the observation process.

4 Payoff Structure

The revised payoff is

$$\Pi_{j,t} = \mu_j K_{j,t} \mathbb{E}_t[\widetilde{M}_{j,t}] x_{j,t} - \frac{\gamma_j}{2} x_{j,t}^2 - \phi_{j,t} x_{j,t} \sum_{k \neq j} x_{k,t}. \quad (21)$$

The three components are:

1. market participation benefit,
2. convex private cost of impersonation,
3. bilateral interference cost.

Define

$$b_{j,t} = \mu_j K_{j,t} \mathbb{E}_t[\widetilde{M}_{j,t}]. \quad (22)$$

Then

$$\Pi_{j,t} = b_{j,t} x_{j,t} - \frac{\gamma_j}{2} x_{j,t}^2 - \phi_{j,t} x_{j,t} \sum_{k \neq j} x_{k,t}. \quad (23)$$

This specification makes the economic source of the original first-order condition explicit.

5 Best Responses

The derivative with respect to $x_{j,t}$ is

$$\frac{\partial \Pi_{j,t}}{\partial x_{j,t}} = b_{j,t} - \gamma_j x_{j,t} - \phi_{j,t} \sum_{k \neq j} x_{k,t}. \quad (24)$$

The second derivative is

$$\frac{\partial^2 \Pi_{j,t}}{\partial x_{j,t}^2} = -\gamma_j < 0. \quad (25)$$

Hence each player's payoff is strictly concave in its own action. The unconstrained best response is

$$BR_{j,t} = \frac{b_{j,t} - \phi_{j,t} \sum_{k \neq j} x_{k,t}}{\gamma_j}. \quad (26)$$

The constrained best response is

$$BR_{j,t}^c = \Pi_{[0,1]} \left[\frac{b_{j,t} - \phi_{j,t} \sum_{k \neq j} x_{k,t}}{\gamma_j} \right], \quad (27)$$

where

$$\Pi_{[0,1]}(z) = \max\{0, \min\{1, z\}\}. \quad (28)$$

6 Matrix Representation

For the homogeneous-interference case,

$$\gamma_j = \gamma, \quad \phi_{j,t} = \phi_t, \quad (29)$$

define

$$W = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}. \quad (30)$$

The equilibrium first-order conditions become

$$[\gamma I + \phi_t W] \mathbf{x}_t^* = \mathbf{b}_t. \quad (31)$$

Define the interference ratio

$$\rho_t = \frac{\phi_t}{\gamma}. \quad (32)$$

Then

$$(I + \rho_t W) \mathbf{x}_t^* = \frac{1}{\gamma} \mathbf{b}_t. \quad (33)$$

The eigenvalues of $I + \rho_t W$ are

$$1 - \rho_t, \quad 1 - \rho_t, \quad 1 + 2\rho_t. \quad (34)$$

Therefore the unconstrained matrix becomes singular at

$$\rho_t = 1 \quad (35)$$

for $\rho_t \geq 0$.

This is a singularity of the interior linear system. It should not automatically be interpreted as economic collapse.

7 Closed-Form Interior Equilibrium

For

$$\rho_t \neq 1, \quad \rho_t \neq -\frac{1}{2}, \quad (36)$$

the inverse exists and

$$(I + \rho_t W)^{-1} = \frac{1}{1 - \rho_t} I - \frac{\rho_t}{(1 - \rho_t)(1 + 2\rho_t)} \mathbf{1}\mathbf{1}^\top. \quad (37)$$

Hence

$$\mathbf{x}_t^* = \frac{1}{\gamma} (I + \rho_t W)^{-1} \mathbf{b}_t. \quad (38)$$

For agent j ,

$$x_{j,t}^* = \frac{1}{\gamma} \frac{b_{j,t}(1 + \rho_t) - \rho_t \sum_{k \neq j} b_{k,t}}{(1 - \rho_t)(1 + 2\rho_t)}. \quad (39)$$

This expression describes the interior solution only.

The actual equilibrium must satisfy

$$x_{j,t}^* \in [0, 1]. \quad (40)$$

When an interior solution violates these constraints, the equilibrium must be determined using the constrained optimization conditions.

8 Constrained Nash Equilibrium

For each agent define the lower and upper multipliers

$$\lambda_{j,t}^L \geq 0, \quad \lambda_{j,t}^U \geq 0. \quad (41)$$

The KKT conditions are

$$b_{j,t} - \gamma x_{j,t}^* - \phi_t \sum_{k \neq j} x_{k,t}^* + \lambda_{j,t}^L - \lambda_{j,t}^U = 0, \quad (42)$$

with

$$x_{j,t}^* \geq 0, \quad 1 - x_{j,t}^* \geq 0, \quad (43)$$

and complementary slackness:

$$\lambda_{j,t}^L x_{j,t}^* = 0, \quad (44)$$

$$\lambda_{j,t}^U (1 - x_{j,t}^*) = 0. \quad (45)$$

This formulation correctly distinguishes an interior equilibrium from corner solutions.

9 Potential Game Representation

The game admits the potential function

$$\mathcal{P}(\mathbf{x}_t) = \mathbf{b}_t^\top \mathbf{x}_t - \frac{\gamma}{2} \mathbf{x}_t^\top \mathbf{x}_t - \phi_t \sum_{j < k} x_{j,t} x_{k,t}. \quad (46)$$

Equivalently,

$$\mathcal{P}(\mathbf{x}_t) = \mathbf{b}_t^\top \mathbf{x}_t - \frac{1}{2} \mathbf{x}_t^\top (\gamma I + \phi_t W) \mathbf{x}_t. \quad (47)$$

The Hessian of $-\mathcal{P}$ is

$$H = \gamma I + \phi_t W. \quad (48)$$

For $\phi_t \geq 0$, positive definiteness requires

$$\gamma - \phi_t > 0, \quad (49)$$

or equivalently,

$$\rho_t < 1. \quad (50)$$

Thus the mathematically meaningful regularity condition is

$$0 \leq \rho_t < 1. \quad (51)$$

The point $\rho_t = 1$ is a loss of strict convexity of the potential problem, rather than a universal claim that all constrained actions instantaneously become zero.

10 Symmetric Equilibrium

Suppose

$$b_{D,t} = b_{I,t} = b_{A,t} = b_t. \quad (52)$$

Then

$$x_{D,t}^* = x_{I,t}^* = x_{A,t}^* = x_t^*. \quad (53)$$

The first-order condition becomes

$$b_t - \gamma x_t^* - 2\phi_t x_t^* = 0. \quad (54)$$

Therefore

$$\boxed{x_t^* = \frac{b_t}{\gamma + 2\phi_t}} \quad (55)$$

or

$$\boxed{x_t^* = \frac{b_t/\gamma}{1 + 2\rho_t}}. \quad (56)$$

Consequently,

$$\frac{\partial x_t^*}{\partial \rho_t} = -\frac{2b_t/\gamma}{(1 + 2\rho_t)^2} < 0. \quad (57)$$

Furthermore,

$$\lim_{\rho_t \rightarrow 1^-} x_t^* = \frac{b_t}{3\gamma}, \quad (58)$$

whereas

$$\lim_{\rho_t \rightarrow \infty} x_t^* = 0. \quad (59)$$

Thus the symmetric equilibrium does not collapse to zero at $\rho_t = 1$. The value $\rho_t = 1$ is instead the singularity of the heterogeneous interior linear system and the boundary of strict convexity of the potential formulation.

11 Strategic Substitutability

For $j \neq k$,

$$\frac{\partial BR_j}{\partial x_k} = -\frac{\phi_t}{\gamma} = -\rho_t. \quad (60)$$

Therefore, whenever

$$\phi_t > 0, \quad (61)$$

the actions are strategic substitutes.

The direct knowledge effect is

$$\frac{\partial x_{j,t}^*}{\partial K_{j,t}} > 0 \quad (62)$$

in the interior region.

The cross-knowledge effect is

$$\frac{\partial x_{j,t}^*}{\partial K_{k,t}} < 0, \quad k \neq j, \quad (63)$$

when $\phi_t > 0$ and the equilibrium remains interior.

Thus knowledge can have both direct and strategic effects.

12 The Knowledge Wedge

The adversarial knowledge wedge is defined as

$$\mathcal{W}_{D,t} = x_{D,t}^*(K_{D,t}, K_{I,t}, K_{A,t}) - x_{D,t}^*(K_{D,t}, K_{I,t}, 0). \quad (64)$$

Under positive interference,

$$\mathcal{W}_{D,t} \leq 0 \quad (65)$$

in the interior region whenever the cross-knowledge effect is negative.

This provides an empirical definition of adversarial crowding-out.

13 Dynamic MAID Model

The static knowledge state can be replaced by a stochastic process:

$$K_{j,t+1} = K_{j,t} + \alpha_{K,j} (K_j^* - K_{j,t}) + \sigma_{K,j} \xi_{j,t+1}, \quad (66)$$

subject to

$$0 \leq K_{j,t} \leq 1. \quad (67)$$

The interference ratio may also evolve:

$$\rho_{t+1} = \rho_t + \alpha_\rho (\rho^* - \rho_t) + \sigma_\rho \zeta_{t+1}. \quad (68)$$

The resulting strategic state is

$$Z_t = \begin{pmatrix} K_{D,t} \\ K_{I,t} \\ K_{A,t} \\ \rho_t \end{pmatrix}. \quad (69)$$

The equilibrium action is then

$$\mathbf{x}_t^* = \mathcal{M}(Z_t, X_t). \quad (70)$$

14 Strategic Observation Distortion

The central empirical extension is to allow strategic actions to affect observed market variables.

Let the latent economic state be

$$X_t. \quad (71)$$

The conventional measurement equation is

$$Y_t^{obs} = HX_t + \epsilon_t. \quad (72)$$

The MAID measurement equation becomes

$$\boxed{Y_t^{obs} = HX_t + G\mathbf{x}_t^* + \epsilon_t.} \quad (73)$$

Here:

$$H = \text{fundamental measurement matrix}, \quad (74)$$

$$G = \text{strategic-distortion matrix}, \quad (75)$$

and

$$\epsilon_t \sim N(0, R_t). \quad (76)$$

This decomposition distinguishes

$$\underbrace{HX_t}_{\text{fundamental signal}} + \underbrace{G\mathbf{x}_t^*}_{\text{strategic distortion}} + \underbrace{\epsilon_t}_{\text{measurement noise}}. \quad (77)$$

15 State-Space Form

Define the augmented state

$$\mathcal{X}_t = \begin{pmatrix} X_t \\ K_{D,t} \\ K_{I,t} \\ K_{A,t} \\ \rho_t \end{pmatrix}. \quad (78)$$

The transition equation is

$$\mathcal{X}_{t+1} = F(\mathcal{X}_t) + \Omega\eta_{t+1}. \quad (79)$$

The observation equation is

$$Y_t^{obs} = HX_t + G\mathcal{M}(\mathcal{X}_t) + \epsilon_t. \quad (80)$$

This is generally nonlinear because the equilibrium mapping $\mathcal{M}$ is nonlinear in the underlying parameters.

The appropriate estimator is therefore an extended Kalman filter, unscented Kalman filter, particle filter, or simulated likelihood procedure, depending on the specification.

16 Time-Varying Measurement Uncertainty

The original model treats uncertainty largely through the knowledge variable. The improved model allows knowledge to influence the measurement covariance.

Define

$$R_t = R_0 + R_K(1 - \bar{K}_t) + R_\rho \rho_t^2, \quad (81)$$

where

$$\bar{K}_t = \frac{1}{3} \sum_{j \in J} K_{j,t}. \quad (82)$$

Then

$$\bar{K}_t \downarrow \implies R_t \uparrow. \quad (83)$$

The interpretation is that observations become statistically less reliable when epistemic uncertainty increases.

This produces a distinction between:

$$\text{fundamental volatility} \quad (84)$$

and

$$\text{epistemic measurement uncertainty}. \quad (85)$$

17 Bayesian Updating of Knowledge

Let

$$K_{j,t} \sim \text{Beta}(\alpha_{j,t}, \beta_{j,t}). \quad (86)$$

After observing $s_{j,t}$ correct predictions out of $n_{j,t}$ observations,

$$K_{j,t} \mid s_{j,t}, n_{j,t} \sim \text{Beta}(\alpha_{j,t} + s_{j,t}, \beta_{j,t} + n_{j,t} - s_{j,t}). \quad (87)$$

The posterior mean is

$$\hat{K}_{j,t} = \frac{\alpha_{j,t} + s_{j,t}}{\alpha_{j,t} + \beta_{j,t} + n_{j,t}}. \quad (88)$$

The posterior variance is

$$\text{Var}(K_{j,t} \mid \mathcal{F}_t) = \frac{\alpha_{j,t}^+ \beta_{j,t}^+}{(\alpha_{j,t}^+ + \beta_{j,t}^+)^2 (\alpha_{j,t}^+ + \beta_{j,t}^+ + 1)}, \quad (89)$$

where

$$\alpha_{j,t}^+ = \alpha_{j,t} + s_{j,t}, \quad (90)$$

and

$$\beta_{j,t}^+ = \beta_{j,t} + n_{j,t} - s_{j,t}. \quad (91)$$

18 Variance Propagation

Let

$$\boldsymbol{\theta}_t = \begin{pmatrix} K_{D,t} \\ K_{I,t} \\ K_{A,t} \end{pmatrix}. \quad (92)$$

For the equilibrium mapping

$$\mathbf{x}_t^* = \mathcal{M}(\boldsymbol{\theta}_t, \rho_t), \quad (93)$$

a first-order delta-method approximation gives

$$\text{Var}(\mathbf{x}_t^*) \approx J_t \text{Var}(\boldsymbol{\theta}_t) J_t^\top, \quad (94)$$

where

$$J_t = \left. \frac{\partial \mathcal{M}}{\partial \boldsymbol{\theta}} \right|_{\boldsymbol{\theta}_t}. \quad (95)$$

The uncertainty generated by imperfect knowledge is therefore propagated into equilibrium strategic actions.

19 Empirical Identification

The model produces four classes of observable quantities:

$$Y_t^{obs}, \quad \hat{X}_t, \quad \hat{K}_t, \quad \hat{\rho}_t. \quad (96)$$

A generalized method of moments estimator may be constructed from

$$e_{j,t} = x_{j,t}^{obs} - x_{j,t}^*(\boldsymbol{\theta}), \quad (97)$$

with moment condition

$$\mathbb{E}[Z_t e_{j,t}] = 0. \quad (98)$$

The GMM estimator is

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} g_T(\boldsymbol{\theta})^\top W_T g_T(\boldsymbol{\theta}), \quad (99)$$

where

$$g_T(\boldsymbol{\theta}) = \frac{1}{T} \sum_{t=1}^T Z_t e_t(\boldsymbol{\theta}). \quad (100)$$

20 Financial-Market Application

For a yield-curve application, let

$$Y_t = \begin{pmatrix} y_{2,t} \\ y_{3,t} \\ y_{5,t} \\ y_{7,t} \\ y_{10,t} \\ y_{20,t} \\ y_{30,t} \end{pmatrix}. \quad (101)$$

Let the latent term-structure state be

$$X_t = \begin{pmatrix} X_{1,t} \\ X_{2,t} \\ X_{3,t} \end{pmatrix}, \quad (102)$$

where the factors may subsequently be identified with real-rate, inflation-risk-premium, and liquidity components.

The MAID-augmented yield equation is

$$\boxed{Y_t = a + BX_t + G\mathbf{x}_t^* + \epsilon_t.} \quad (103)$$

The strategic component is

$$\mathbf{x}_t^* = \mathcal{M}(K_{D,t}, K_{I,t}, K_{A,t}, \rho_t, X_t). \quad (104)$$

This creates a direct bridge between the term-structure model and the MAID model.

21 Cross-Institutional Measurement

Suppose observations are obtained from four institutional measurement channels:

$$j \in \{RBI, Fed, SNB, BIS\}. \quad (105)$$

Then

$$Y_t^{obs} = \begin{pmatrix} Y_t^{RBI} \\ Y_t^{Fed} \\ Y_t^{SNB} \\ Y_t^{BIS} \end{pmatrix}. \quad (106)$$

The institutional observation equations become

$$Y_t^j = H_j X_t + G_j \mathbf{x}_t^* + \epsilon_{j,t}. \quad (107)$$

This allows the model to distinguish disagreement caused by

$$H_j X_t \quad (108)$$

from disagreement caused by

$$G_j \mathbf{x}_t^* \quad (109)$$

and ordinary measurement noise.

An external market-data source can be incorporated through

$$Y_t^{TV} = H_{TV} X_t + G_{TV} \mathbf{x}_t^* + \epsilon_{TV,t}, \quad (110)$$

provided that maturity, security definition, timestamp, and yield convention are first harmonised.

22 Strategic Distortion Index

Define the cross-source residual

$$e_{j,t} = Y_{j,t}^{obs} - \hat{Y}_t. \quad (111)$$

A simple distortion statistic is

$$\mathcal{D}_t = \frac{1}{N_t} \sum_{j=1}^{N_t} \|e_{j,t}\|^2. \quad (112)$$

A covariance-adjusted version is

$$\mathcal{D}_t^M = \frac{1}{N_t} \sum_{j=1}^{N_t} e_{j,t}^\top R_{j,t}^{-1} e_{j,t}. \quad (113)$$

The second statistic distinguishes large discrepancies from discrepancies that are statistically expected given measurement uncertainty.

23 Regime Classification

Define the augmented systemic state

$$S_t = \begin{pmatrix} X_t \\ \mathbf{K}_t \\ \rho_t \\ \mathbf{x}_t^* \\ \mathcal{D}_t \end{pmatrix}. \quad (114)$$

A regime variable may then be defined as

$$R_t = \mathcal{R}(S_t). \quad (115)$$

For example, a high-interference regime may satisfy

$$\rho_t \geq \rho_H, \quad (116)$$

while an epistemically uncertain regime may satisfy

$$\overline{K}_t \leq K_L. \quad (117)$$

A strategic-information stress regime may therefore satisfy

$$R_t^{stress} = 1 \quad (118)$$

if

$$\rho_t \geq \rho_H, \quad \overline{K}_t \leq K_L, \quad \mathcal{D}_t \geq D_H. \quad (119)$$

These thresholds should be estimated or calibrated rather than imposed without empirical justification.

24 Multi-Agent Reinforcement Learning Extension

In a repeated game, agent j has state

$$s_t = (X_t, \mathbf{K}_t, \rho_t, \mathbf{x}_t). \quad (120)$$

The action is

$$a_{j,t} = x_{j,t}. \quad (121)$$

The Bellman equation is

$$Q_j(s_t, a_{j,t}) = \Pi_{j,t} + \beta \mathbb{E}_t \left[\max_{a_{j,t+1}} Q_j(s_{t+1}, a_{j,t+1}) \right]. \quad (122)$$

The static Nash equilibrium becomes a benchmark fixed point:

$$\mathbf{x}_t^* = BR(\mathbf{x}_t^*; X_t, \mathbf{K}_t, \rho_t). \quad (123)$$

The dynamic learning problem therefore asks whether agents converge toward, depart from, or repeatedly cycle around the static equilibrium.

25 Robustness and Adversarial Learning

Let the adversarial agent have action

$$x_{A,t} \in [0, 1]. \quad (124)$$

A robust-control formulation is

$$\max_{x_D} \min_{x_A} \mathbb{E} \left[\sum_{t=0}^T \beta^t \Pi_{D,t} \right]. \quad (125)$$

Alternatively, the adversary may maximize distortion:

$$\max_{x_A} \mathcal{D}_t \quad (126)$$

subject to

$$0 \leq x_{A,t} \leq 1. \quad (127)$$

This makes the connection between strategic impersonation and adversarial machine learning explicit.

26 Welfare and Mechanism Design

Aggregate welfare is

$$W(\mathbf{x}) = \sum_{j \in J} \Pi_j(\mathbf{x}). \quad (128)$$

The social planner solves

$$\mathbf{x}^{SO} = \arg \max_{\mathbf{x} \in [0,1]^3} W(\mathbf{x}). \quad (129)$$

The external interference imposed by agent j on the other agents is

$$MC_j = \phi \sum_{k \neq j} x_k. \quad (130)$$

A Pigouvian mechanism may therefore use the tax

$$\tau_j = \phi \sum_{k \neq j} x_k. \quad (131)$$

The decentralized first-order condition becomes

$$b_j - (\gamma + \tau_j)x_j - \phi \sum_{k \neq j} x_k = 0. \quad (132)$$

The welfare mechanism should be evaluated empirically rather than assuming that the static tax necessarily implements the dynamic optimum.

27 Identification Strategy

The empirical implementation proceeds in five stages.

27.1 Stage 1: Fundamental state estimation

Estimate

$$X_t \quad (133)$$

from financial and macroeconomic observations using a state-space model.

27.2 Stage 2: Knowledge estimation

Estimate

$$\hat{K}_{j,t} \quad (134)$$

from prediction accuracy, information quality, forecast errors, or other observable proxies.

27.3 Stage 3: Strategic estimation

Estimate

$$\hat{\rho}_t = \frac{\hat{\phi}_t}{\hat{\gamma}_t}. \quad (135)$$

27.4 Stage 4: Observation-distortion estimation

Estimate

$$\hat{G} \quad (136)$$

from cross-source discrepancies after controlling for the latent economic state.

27.5 Stage 5: Joint estimation

Estimate

$$\Theta = \{H, G, \gamma, \phi, \mu, \alpha_K, \sigma_K, \alpha_\rho, \sigma_\rho\} \quad (137)$$

using maximum likelihood, simulated likelihood, Bayesian estimation, or GMM depending on observability.

28 Testable Hypotheses

The improved MAID model produces several empirical hypotheses.

28.1 Hypothesis 1: Strategic substitutability

For $j \neq k$,

$$\frac{\partial BR_j}{\partial x_k} < 0. \quad (138)$$

28.2 Hypothesis 2: Knowledge advantage

Holding other variables constant,

$$\frac{\partial x_j^*}{\partial K_j} > 0. \quad (139)$$

28.3 Hypothesis 3: Adversarial knowledge wedge

Under positive interference,

$$\frac{\partial x_j^*}{\partial K_k} < 0, \quad k \neq j. \quad (140)$$

28.4 Hypothesis 4: Epistemic amplification

As interference increases,

$$\text{Var}(x_j^*) \quad (141)$$

should increase when uncertainty in the knowledge states propagates through the equilibrium mapping.

28.5 Hypothesis 5: Strategic measurement distortion

If strategic actions influence observations,

$$\text{Cov}(\mathcal{D}_t, \rho_t) > 0 \quad (142)$$

after controlling for fundamental volatility and ordinary measurement error.

29 Limitations

The MAID model does not establish that real financial observations are actually generated by impersonation. The strategic-distortion channel is a structural hypothesis that requires empirical identification.

Likewise, fuzzy knowledge $K_{j,t}$ is not automatically observable. Empirical implementation therefore requires measurable proxies or a joint latent-variable estimation procedure.

Finally, the singularity at

$$\rho = 1 \quad (143)$$

should not be described as a universal economic collapse threshold. It is a mathematical boundary of the strictly convex potential formulation and a singularity of the unconstrained heterogeneous interior system. Economic regime changes must be established separately.

30 Conclusion

The improved MAID model separates four concepts that were previously partially conflated:

$$\boxed{\text{fundamental uncertainty} \neq \text{epistemic uncertainty} \neq \text{strategic distortion} \neq \text{measurement noise.}} \quad (144)$$

The resulting observation equation is

$$\boxed{Y_t^{obs} = HX_t + G\mathbf{x}_t^* + \epsilon_t.} \quad (145)$$

The equilibrium strategic state is

$$\boxed{\mathbf{x}_t^* = \mathcal{M}(X_t, \mathbf{K}_t, \rho_t).} \quad (146)$$

The augmented systemic state is

$$\boxed{S_t = (X_t, \mathbf{K}_t, \rho_t, \mathbf{x}_t^*, \mathcal{D}_t).} \quad (147)$$

This formulation preserves the original MAID insight while making the model mathematically internally consistent and directly compatible with state-space estimation, financial-market data, adversarial learning, and multi-agent reinforcement learning.

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